6.1. Random variable

Introduction:

Let the random experiment be the toss of a coin. When 'n' coins are tossed, one may be interested in knowing the number of heads obtained. When a pair of dice is thrown, one may seek information about the sum of sample points. Thus, we associate a real number with each outcome of an experiment. In other words, we are considering a function whose domain is the set of possible outcomes and whose range is subset of the set of real numbers. Such a function is called random variable.

In algebra, you learned about different variables like X or Y or any other letter in a particular problem. Thus in basic mathematics, a variable is an alphabetical character that represents an unknown number. A random variable is a variable that is subject to randomness, which means it could take on different values. In statistics, it is quite general to use X to denote a random variable and it takes on different values depending on the situation.

Some of the examples of random variable:

(i) Number of heads, if a coin is tossed 8 times.

(ii) The return on an investment in one-year period.

(iii) Faces on rolling a die.

(iv) Number of customers who arrive at a bank in the regular interval of one hour between 9.00 a.m and 4.30 p.m from Monday to Friday.

(v) The sale volume of a store on a particular day.

For instance, the random experiment 'E' consists of three tosses of a coin and the outcomes of this experiment forms the sample space is 'S'. Let X denotes the number of heads obtained.



Tossing a Coin
Fig.6.1

Here X is a real number connected with the outcome of a random experiment E. The details given below

Outcome (ω)	Values of $X=x$
(HHH)	3
(HHT)	2
(HTH)	2
(THH)	2

Outcome (ω)	Values of $X=x$
(HTT)	1
(THT)	1
(TTH)	1
(TTT)	0

i.e., $R_X = \{0, 1, 2, 3\}$

From the above said example, for each outcomes ω, there corresponds a real number $X(\omega)$. Since the points of the sample space 'S' corresponds to outcomes is defined for each $\omega \in S$.

6.1.1 Definition of a random variable

Definition 6.1

A random variable (r.v.) is a real valued function defined on a sample space S and taking values in $(-\infty, \infty)$ or whose possible values are numerical outcomes of a random experiment.

Note

(i) If x is a real number, the set of all ω in S such that $X(\omega)=x$ is, denoted by $X=x$. Thus $P(X=x) = P\{\omega:X(\omega)=x\}$.

(ii) $P(X \leq a) = P\{\omega:X(\omega) \in (-\infty, a]\}$ and $P(a < X \leq b) = P\{\omega:X(\omega) \in (a, b]\}$.

(iii) One-dimensional random variables will be denoted by capital letters, $X, Y, Z, \ldots$, etc. A typical outcome of the experiment will be denoted by ω. Thus $b \in X(\omega)$ represents the real number which the random variable X associates with the outcome τ. The values which $X, Y, Z, \ldots$, etc, can assume are denoted by lower case letters, *viz.*, $x, y, z, \ldots$, etc.

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